

The Jacobian matrix is the transposed of the gradient matrix. In the gradient matrix, each row corresponds to a variable, and each column to a function. For the Jacobian matrix, it is the opposite. Here, as we have 2 variables and 3 functions, we have

$$\nabla f(\mathbf{x}) \in \mathbb{R}^{2 \times 3}, \quad J(\mathbf{x}) \in \mathbb{R}^{3 \times 2}.$$

The gradient matrix is

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & \frac{\partial f_3}{\partial x_1} \\ \frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_3}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 2x_1x_2 & -\sin(e^{x_1+x_2^2})e^{x_1+x_2^2} & 0 \\ x_1^2 & -2x_2 \sin(e^{x_1+x_2^2})e^{x_1+x_2^2} & 1 \end{pmatrix}.$$

The Jacobian matrix is

$$J(\mathbf{x}) = \nabla f(\mathbf{x})^T = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 2x_1x_2 & x_1^2 \\ -\sin(e^{x_1+x_2^2})e^{x_1+x_2^2} & -2x_2 \sin(e^{x_1+x_2^2})e^{x_1+x_2^2} \\ 0 & 1 \end{pmatrix}.$$